T3 D_-2: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

D_-2

Additional quartics: 5, additional bilinears: 1, extra Yukawa: y2Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{47{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{5{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} &= 4\mathrm{muT3GeneralS2Sq}\,(\mathrm{lambdaT3GeneralQ001102I1})^* \\ &+ 4\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{muT3GeneralS2Sq})^* \\ &- 12{g_2}^2 m_1^2 \\ &- 6{g_1}^2 m_1^2 \\ &+ 2\lambda_{12}^{(1)} m_2^2 \\ &+ m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} &= 12\mathrm{muT3GeneralS2Sq}\,(\mathrm{lambdaT3GeneralQ000013I1})^* \\ &+ 12\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{muT3GeneralS2Sq})^* \\ &+ 6\lambda_{12}^{(1)} m_1^2 \\ &+ 4\lambda_{S_2}^{(1)} m_2^2\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} &= \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ &- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ &- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ &- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ &+ 4\lambda_5\,(\lambda_5)^* \\ &+ 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\ &+ 8\mathrm{lambdaT3GeneralQ110002I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^* \\ &+ \frac{9{g_2}^4}{4} \\ &+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ &- 9{g_2}^2\lambda_1\end{aligned}$$

$$
\begin{aligned}
&+ \frac{3{g_1}^4}{4} \\
&+ 12{\lambda_1}^2 \\
&+ 6{\lambda_{HS_1}^{(1)}}^2 \\
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 2{\lambda_{HS_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & 8\text{lambdaT3GeneralQ001102I1 } (\text{lambdaT3GeneralQ001102I1})^* \\
&+ 22{g_2}^4 \\
&+ {g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_1}^{(1)}\right) \\
&- 24{g_2}^2\lambda_{S_1}^{(1)} \\
&+ 12{g_1}^4 \\
&+ 2{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 14{\lambda_{S_1}^{(1)}}^2 \\
&+ 8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & 288\text{lambdaT3GeneralQ000004I1 } (\text{lambdaT3GeneralQ000004I1})^* \\
&+ 108\text{lambdaT3GeneralQ000013I1 } (\text{lambdaT3GeneralQ000013I1})^* \\
&+ 12\text{lambdaT3GeneralQ001102I1 } (\text{lambdaT3GeneralQ001102I1})^* \\
&+ 8\text{lambdaT3GeneralQ110002I1 } (\text{lambdaT3GeneralQ110002I1})^* \\
&+ 6{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 10{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda_{S_1}^{(A)}) \\
&+ 4\lambda_5\ (\lambda_5)^* \\
&+ 4\text{lambdaT3GeneralQ110002I1 } (\text{lambdaT3GeneralQ001102I1})^* \\
&+ 4\text{lambdaT3GeneralQ001102I1 } (\text{lambdaT3GeneralQ110002I1})^* \\
&+ 4\text{lambdaT3GeneralQ020101I1 } (\text{lambdaT3GeneralQ020101I1})^* \\
&+ 6{g_2}^4
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3{g_1}^4 \\
&- \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6{\lambda_{HS_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ {(Y_d)}^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ {(Y_e)}^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ {(Y_u)}^\dagger\Big)) \\
& + 12\lambda_5\,(\lambda_5)^* \\
& + 12\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^* \\
& + 4\mathrm{lambdaT3GeneralQ110002I1}(3\,(\mathrm{lambdaT3GeneralQ000013I1})^* + 4\,(\mathrm{lambdaT3GeneralQ110002I1})^*) \\
& + 12\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\
& - \frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
& - \frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
& + 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
& + 6\lambda_1\lambda_{HS_2}^{(1)} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & 4\lambda_5\,(\lambda_5)^* \\
& + 12\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{lambdaT3GeneralQ001102I1})^* \\
& + 4\mathrm{lambdaT3GeneralQ001102I1}(3\,(\mathrm{lambdaT3GeneralQ000013I1})^* + 4\,(\mathrm{lambdaT3GeneralQ001102I1})^*) \\
& + 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\
& - 12{g_2}^2\lambda_{12}^{(1)} \\
& - 6{g_1}^2\lambda_{12}^{(1)} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
& + 4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)} \\
& + 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$\begin{aligned}16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) - 4\lambda_{HS_1}^{(A)}) \\ &+ 4\mathrm{lambdaT3GeneralQ001102I1}\ (\mathrm{lambdaT3GeneralQ020101I1})^* \\ &+ 8\mathrm{lambdaT3GeneralQ110002I1}\ (\mathrm{lambdaT3GeneralQ020101I1})^* \\ &- \frac{21{g_2}^2\lambda_5}{2} \\ &- \frac{9{g_1}^2\lambda_5}{2} \\ &+ 2\lambda_{12}^{(1)}\lambda_5 \\ &+ 2\lambda_1\lambda_5 \\ &+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\ &+ 4\lambda_{HS_2}^{(1)}\lambda_5\end{aligned}$$

$\lambda_{HS_1}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = \lambda_{HS_1}^{(A)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)}) \\ &- \frac{8}{3}\lambda_5\ (\lambda_5)^* \\ &- \frac{8}{3}\mathrm{lambdaT3GeneralQ020101I1}\ (\mathrm{lambdaT3GeneralQ020101I1})^* \\ &+ {g_1}^2\left(-4{g_2}^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right) \\ &- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2} \\ &+ 2\lambda_1\lambda_{HS_1}^{(A)} \\ &+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\ &+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}\end{aligned}$$

$\lambda_{S_1}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = &-2{g_2}^4 \\ &+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right) \\ &- 24{g_2}^2\lambda_{S_1}^{(A)} \\ &+ 3{\lambda_{HS_1}^{(A)}}^2 \\ &+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\ &+ 20{\lambda_{S_1}^{(A)}}^2\end{aligned}$$

$lambdaT3GeneralQ000004I1$

$$16\pi^2 \frac{dlambdaT3GeneralQ000004I1}{d\ln\mu} = 12\lambda_{S_2}^{(1)}\mathrm{lambdaT3GeneralQ000004I1}$$

$$
\begin{aligned}
&+ 9\text{lambdaT3GeneralQ000013I1}^2 \\
&+ 3\text{lambdaT3GeneralQ001102I1}^2 \\
&+ 2\text{lambdaT3GeneralQ110002I1}^2
\end{aligned}
$$

$lambdaT3GeneralQ000013I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000013I1}{d\ln\mu} = {} & 72\text{lambdaT3GeneralQ000004I1 } (\text{lambdaT3GeneralQ000013I1})^* \\
& + 6\lambda_{12}^{(1)}\text{lambdaT3GeneralQ001102I1} \\
& + 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ110002I1} \\
& + 18\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ000013I1}
\end{aligned}
$$

$lambdaT3GeneralQ001102I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ001102I1}{d\ln\mu} = {} & 24\text{lambdaT3GeneralQ000004I1 } (\text{lambdaT3GeneralQ001102I1})^* \\
& - 12{g_2}^2\text{lambdaT3GeneralQ001102I1} \\
& - 6{g_1}^2\text{lambdaT3GeneralQ001102I1} \\
& + \lambda_{12}^{(1)}(6\text{lambdaT3GeneralQ000013I1} + 8\text{lambdaT3GeneralQ001102I1}) \\
& + 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ110002I1} \\
& + 8\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ001102I1} \\
& + 4\lambda_{S_1}^{(A)}\text{lambdaT3GeneralQ001102I1} \\
& + 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ001102I1} \\
& + 4\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ020101I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ020101I1}{d\ln\mu} = {} & \text{lambdaT3GeneralQ020101I1}(6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& + 4\text{lambdaT3GeneralQ001102I1 } (\lambda_5)^* \\
& + 8\text{lambdaT3GeneralQ110002I1 } (\lambda_5)^* \\
& - \frac{21{g_2}^2\text{lambdaT3GeneralQ020101I1}}{2} \\
& - \frac{9{g_1}^2\text{lambdaT3GeneralQ020101I1}}{2} \\
& + 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
& + 2\lambda_1\text{lambdaT3GeneralQ020101I1} \\
& + 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
& - 4\lambda_{HS_1}^{(A)}\text{lambdaT3GeneralQ020101I1} \\
& + 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ110002I1$

$$
16\pi^2 \frac{dlambdaT3GeneralQ110002I1}{d\ln\mu} = \text{lambdaT3GeneralQ110002I1}(6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))
$$

$$
\begin{aligned}
&+ 24\text{lambdaT3GeneralQ000004I1}\,(\text{lambdaT3GeneralQ110002I1})^* \\
&- \frac{9{g_2}^2\text{lambdaT3GeneralQ110002I1}}{2} \\
&- \frac{3{g_1}^2\text{lambdaT3GeneralQ110002I1}}{2} \\
&+ 6\lambda_1\text{lambdaT3GeneralQ110002I1} \\
&+ 6\lambda^{(1)}_{HS_1}\text{lambdaT3GeneralQ001102I1} \\
&+ \lambda^{(1)}_{HS_2}(6\text{lambdaT3GeneralQ000013I1} + 8\text{lambdaT3GeneralQ110002I1}) \\
&+ 2\lambda^{(1)}_{S_2}\text{lambdaT3GeneralQ110002I1} \\
&+ 12\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$muT3GeneralS2Sq$

$$
\begin{aligned}
16\pi^2\,\frac{dmuT3GeneralS2Sq}{d\ln\mu} = {}& 24\text{lambdaT3GeneralQ000004I1}\,(\text{muT3GeneralS2Sq})^* \\
&+ 2\lambda^{(1)}_{S_2}\text{muT3GeneralS2Sq} \\
&+ 6\text{lambdaT3GeneralQ000013I1}m_2^2 \\
&+ 6\text{lambdaT3GeneralQ001102I1}m_1^2
\end{aligned}
$$